



FNB

APPLICATION PROGRAMMING INTERFACE (API) TRANSACTION HISTORY PRODUCT CONFIGURATION MANUAL

SOUTH AFRICA



TRANSACTION HISTORY API

DESCRIPTION

This document provides channel guidelines specific to retrieving Transaction History on APIs.

This document details the transports available, message formats and channel features available on Integration Channel for Transaction History.

This document further outlines what the API can do, what products are supported and how to connect to and use the API.

CHANNEL FEATURES

- ✓ Allow/revoke access for certain accounts.
- ✓ Give/revoke consent to Third-Party.

API SCOPE

The API is currently available in South Africa, Lesotho, Zambia, Eswatini, Botswana and covers transact accounts.

This API is available for business, commercial, corporate and investment customers.

WHAT CAN THE API DO?

This service allows the client to:

- Connect to the API on their own behalf.
- Connect to the API through a Third-Party.
- Retrieve a list of transactions including account balances for a defined date range related to the chosen account.

HOW TO GET THE API

You need to be an Online Banking Enterprise™ user or complete the platform registration.

There are a few ways in which you can get and connect to the API.

- ✓ **Unassisted:** With unassisted, the client can subscribe for the Transaction History API on the Integration Channel, which is found under Business Solutions tab on Online Banking Enterprise™.
- ✓ **Assisted:** The client can contact their Platform Specialist, Transactional Portfolio Manager, or Implementation Manager for assistance.



HOW TO CONNECT TO THE API

The client can connect and consume the API in two ways:

1. On my own behalf

The client can connect to the API directly from their line of business system. This can be achieved without a technology intermediary or Third-Party (System Operator or technology partner).

In both unassisted and assisted journeys, you can maintain your connection details to your line of business system.

2. Through a Third-Party

The client can delegate the API processing and connection responsibility to an intermediary or Third-Party (System Operator or technology partner).

With this connection type, the client will be required to provide the bank with consent to share their product account information with the Third-Party as well as indicate which accounts the Third-Party can retrieve information on.

In both unassisted and assisted journeys, the client can maintain or revoke the consent for the Third-Party to act on your behalf as well as the selection of the accounts.

HOW TO GET THE API

SECURITY AND ACCESS CONTROL

Our APIs are secured and protected. We require positive authentication, authorisation and access tokens to gain access to the API.

- **Authentication**

API client authentication use JWT signed tokens. Authentication will be done through the use of a client ID and client secret that will serve as credentials to positively identify the client. The credentials are provided through the subscribe process on the Integration Channel.

- **Authorisation**

Authorisation is achieved through the OAuth 2.0 standard using the authorisation code flow. The authorisation code flow includes using an Auth Code to receive an access token to initiate the process to make calls. When connecting through a Third-Party, the authorisation can be done in two ways:

- ✓ **Auth Code:** The client receives the Auth Code when subscribing to the service on Integration Channel and share the Auth Code securely with the Third-Party to connect to the API.
- ✓ **OAuth 2.0:** The Third-Party provides their redirect URL when subscribing to the service and the client will be redirected to the Third-Party's website when they complete the subscription and choosing to connect through a Third-Party.



- **Access Token**

This is treated as a subset of authorisation.

Access tokens can be obtained through the OAuth 2.0 token endpoint by either presenting the authorisation code or a refresh token. An access token is used each time a call is made and has a set life span, once expired the refresh token can be used to request a new access token. In the instance where the refresh token expires, the client will need to request for a new access and refresh token by initiating the subscribe process again.

WHAT FORMAT IS THE API IN?

Transaction History API uses OpenAPI Specification (OAS) standard.

OpenAPI Specification (OAS) is an industry standard, programming language-agnostic specification standard for RESTful APIs. OAS allows easy access to discover and understand the API without having access to the source code, documentation or implementation logic. OAS is also widely known as a Swagger Document.

APPENDIX

Below is the referenced information which should be read in conjunction with this document. The information can also be retrieved on Integration Channel, Transaction History catalogue card under the API subtab:

- Transaction History API Message Specification
- TransactionHistoryAPI Swagger Document

API REFERENCE

The Transaction API follows the JSON Format, the API uses a date range to be able to retrieve the transactions within the defined date range and chosen account.

The API is enabled through the below method:

- ✓ **GET – retrieveTransactionHistory**

Retrieve Transaction History:

Allows you to retrieve a list of transactions including account balances for the defined date range related to the chosen account.

Our APIs use polling method, which allows you to query the API at regular intervals to check for new data.

PROCESS FLOWS



Transaction History – On My Own Behalf

SUBSCRIBE

Choose how to connect to the Transaction History API and choose the Account(s) to retrieve transaction history for.

AUTHENTICATION AND AUTHORISATION

Create the required connectivity protocol and receive API credentials (Client ID and Secret), and/or Access Codes if connected via Third-Party.

The credentials will be used to authenticate and authorise the service when making the API call.

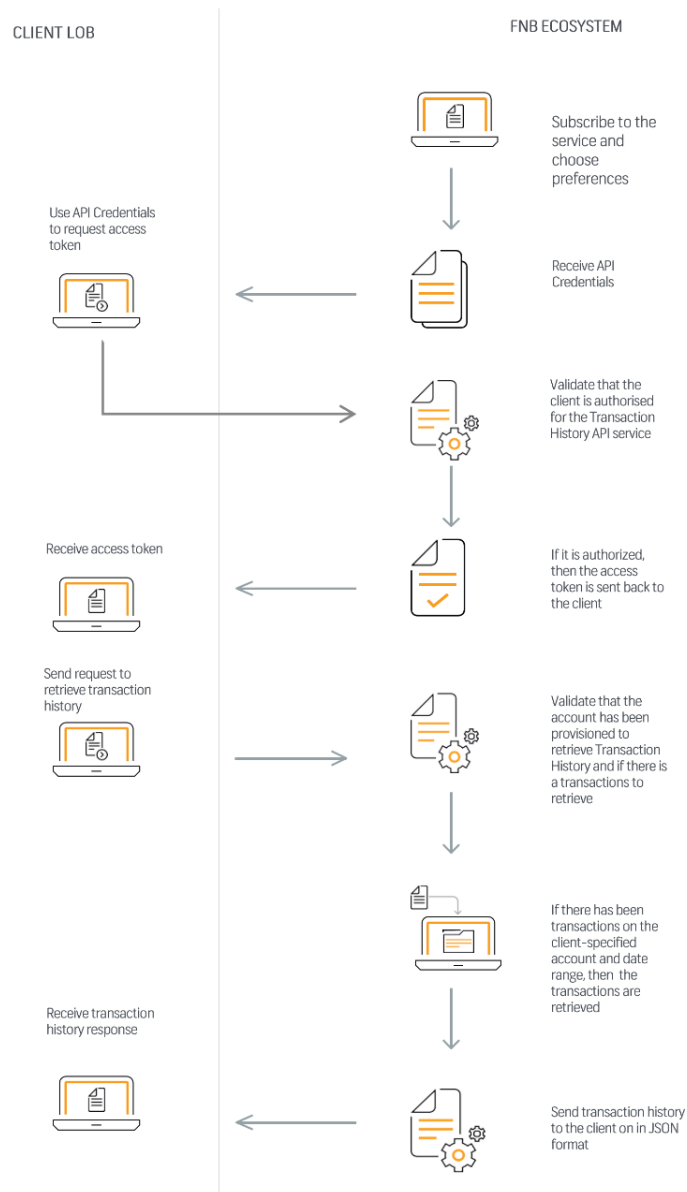
TRANSACTION INITIATION

Choose the account and date range to retrieve transaction history for. If there are available transactions then these can be retrieved.

PROCESSING

Use the API credentials to get authorisation and generate the access token. Send the request through the Transaction History API using the Start and end date and Account. If there are transactions available to be retrieved, then the API service will respond with the transaction list in JSON format .

- A successful response will have HTTP Status code 200.
- If the request has been submitted successfully but there is no available response to be retrieved as yet, then the service will return HTTP status code 204.
- If the access token has expired, then the service will return HTTP status code 401 and a new Token will need to be requested through a refresh token.
- If the service is unavailable or unable to process the request, it will return HTTP Status code 500.





Transaction History– Through a Third-Party

SUBSCRIBE

The Third-Party will first need to connect to the specific services they would like to provide and choose the countries where they will provide these services.

The Third-Party will receive their own API credentials and choose an authorization type.

Two authorisation types exist:

1. **Authorisation code** – Authorisation codes – Auth Code shared by the Client to the Third-Party which will be used when processing.
2. **OAuth2.0** – Redirect URL used for the client to link to the Third-Party and authorise access.

Navigate to Integration Channel and Choose to connect to the Transaction History API "Through a Third-Party" and choose the desired preferences.

Preferences include: Choosing accounts that are allowed to retrieve Transaction History

AUTHENTICATION AND AUTHORISATION

Create the required connectivity protocol and receive API credentials (Client ID and Secret), and/or Access Codes if authorization type is Auth Code.

The credentials will be used to authenticate and authorise the service when making the API call.

PROCESSING

To process transactions, a valid Access token and API credentials will be required, once those have been successfully validated processing can commence.

When processing through the Third-Party, the Third-Party will use their own API credentials to process transactions on behalf of the client and use the Auth Code to link to that specific client.

To process, send the API request with the mandatory fields populated, Channel and Bank validations will be done on the Transaction History request thereafter retrieve the response. The request and response will be in JSON format.

The processing of Transaction History API allows for single and bulk requests:

- A successful response will have HTTP Status code 200.
- If the request has been submitted successfully but there is no available response to be retrieved as yet, then the service will return HTTP status code 204.
- If the access token has expired, then the service will return HTTP status code 401 and a new Token will need to be requested through a refresh token.
- If the service is unavailable or unable to process the request, it will return HTTP Status code 500.

THIRD PARTY LOB

CLIENT LOB

FNB ECOSYSTEM

